

Video Transit Results — Executive Brief

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Executive summary

The prototype delivered every video it was given intact, and the choices it automates would save an estimated **\$29,000–\$141,000 a year** for a platform serving 1 million viewer-hours a month. Two further options, below the total line, could add more.

We tested 15 video transfers between two sites: recorded video, adaptive streaming packages (HLS and DASH) and live feeds, over four simulated network types from office LAN to mobile. All 15 arrived bit-for-bit identical to what was sent. The savings come mainly from sending fewer bytes for the same picture, and from not re-processing live video that is already in a deliverable format.

Lever	What the test showed	Est. savings per month	Per year
Smaller video for mobile and slow connections	39% fewer bytes, picture quality nearly unchanged	\$1,390 – \$6,950	\$16,700 – \$83,400
Newer video format (AV1) for fast-moving content such as sports	22% fewer bytes and visibly better picture	\$790 – \$3,990	\$9,400 – \$47,900
Compressing knowledge-base (UFCS) data syncs	85% less data per sync, 6.7× faster transfer	\$90 – \$390	\$1,000 – \$4,600
Modern image format (WebP)	Images 89% smaller	\$45 – \$225	\$540 – \$2,700
Re-packaging live streams instead of re-encoding them	99% less computing per live hour	\$90 – \$220	\$1,100 – \$2,650
Total for these five levers		\$2,400 – \$11,770	\$28,700 – \$141,200

Lever	What the test showed	Est. savings per month	Per year
Option: share one network link between live and background traffic	Live video 4.3× faster while the link stayed 80% busy with other work	\$810 – \$4,050 per 10 Gbps of capacity	\$9,700 – \$48,600
Option: archive delivery copies instead of studio masters	98% less storage per hour of video	\$3,750 – \$21,580 per 10,000 hours stored	\$45,000 – \$259,000

The low figure uses large-volume contract prices and the high figure uses typical list prices. The workload and every price are listed under *Savings model*, so finance can substitute our real volumes. The engine's own running costs are small by comparison: its framing overhead is about \$90–\$400 a month at this volume, and tamper-proof signing of every packet costs \$20–\$50 per petabyte.

Recommendation: adopt the five core levers for the next platform release. Run a two-week pilot on real traffic to confirm the savings before committing budget. Bring the archive option to the content owners as a policy decision.

What we tested and why it matters

We measured each video at three moments, so every saving traces back to a specific step:

- **Before sending:** how small each video format made the file, and how long the conversion took.
- **In transit:** how much of the network traffic was actual video, and where the time went.
- **After arrival:** whether the copy was perfect, how much computing it takes to play, and how close the picture is to the original.

We used two kinds of video. One is a studio-style clip. The other is a deliberately hard, fast-moving clip that stands in for sports and action. Uncompressed, each runs at 332 Mbps, too large for most links, so the format choice decides what we pay to store and move video.

We ran the transfers over four simulated networks:

Network	Speed	One-way delay	Stands in for
Office LAN	1 Gbps	0.5 ms	Same data centre
Regional	50 Mbps	20 ms	Between nearby regions
Long-haul	20 Mbps	80 ms	Across a continent or ocean
Mobile	3 Mbps	40 ms	A viewer on a phone network

We also tested the three ways video reaches viewers today. HLS (used by Apple devices and most web players) and DASH (used by Android and smart TVs) cover playback. RTMP is how cameras and studio software such as OBS send live video out.

Compression: smaller files, lower delivery and storage bills

At a standard HD setting, every format cut video to 1/100th–1/130th of its raw size with good picture quality. The choice of format and resolution then decides another 20–40% of the delivery bill.

Format	Fast-moving clip: data used	Fast-moving clip: picture score	Studio clip: picture score	Relative cost to encode
H.264 (today's default)	3.2 Mbps, 30% over the 2.5 Mbps target	0.855	0.970	1×
H.265 (HEVC)	3.2 Mbps, 30% over target	0.913	0.970	1.5×
AV1 (newest, royalty-free)	2.5 Mbps, on target	0.917	0.975	1.7×

Picture score is SSIM, an industry measure where 1.0 means identical to the original. Above about 0.95 most viewers see no difference.

What this is worth:

- **AV1 for fast-moving content** saves 0.32 GB per viewer-hour and gives a visibly better picture. That is **\$3–\$16 saved per 1,000 viewer-hours** at contract-to-list delivery prices. Encoding with AV1 costs about 2.4 extra computer-hours per hour of content, roughly

\$0.05–\$0.12, so it repays itself after a few dozen viewers. The caveat is that older phones and TVs cannot play AV1, so it must sit alongside H.264, not replace it.

- **A smaller picture on slow connections** (540p instead of 720p on the 3 Mbps mobile link) saves 0.46 GB per viewer-hour, 39% of the data. That is **\$5–\$23 per 1,000 viewer-hours**, and the picture score drops only from 0.970 to 0.954 on a phone-sized screen. The prototype picks this automatically from the measured connection speed.
- **Knowledge-base data** (the 149-million-record UFCS fact feed) shrinks from 168 GB to 25 GB per full copy. Each sync saves **\$3–\$13** in cross-region transfer fees, and the copy takes about 30 minutes instead of 3.7 hours on a 100 Mbps link.
- **Images** shrink 89%, from 491 KB to 53 KB for an HD photo. That is **\$4.50–\$22 saved per million images** delivered.

Transport efficiency: paying for video, not overhead

Over 99% of the bytes the engine sends are video. Its own labelling, error checks and receipts add only 0.1–1% to the bill.

- **Labelling and error checks** (each packet's address label, content description and checksum) take 0.4% of the traffic for recorded video, and up to 0.9% for live video, which travels in smaller pieces. At 1 million viewer-hours a month that is about 4.5 TB, or **\$90–\$400 a month** in transfer fees. It is a cost, but a small one, and it buys automatic detection and recovery of damaged data.
- **Delivery receipts** sent back by the receiving site add another 0.05–0.3%.
- **Tamper-proof signing** of every packet, which proves where data came from, costs **\$20–\$50 in computing per petabyte**. That is negligible, so we recommend turning it on by default.
- **Use of paid capacity:** on the regional, long-haul and mobile links the engine kept 89–95% of the paid bandwidth busy during transfers. On the office LAN it reached 559 Mbps of 1 Gbps. That figure is limited by our test equipment, not the engine, so it understates real LAN speed.

When we deliberately damaged data in earlier tests, the engine lost only the damaged piece and carried on. A corrupted stream does not have to be resent from the start, which avoids repeat transfer fees and viewer-visible stalls.

Decompression and processing: low computing cost

Unpacking and checking video on arrival costs cents per thousand hours. The real computing saving is in live video: re-packaging it instead of re-encoding it cuts the work by 99%.

- **Checking each packet on arrival** takes about 30 millionths of a second, a rate of 1.2–2.9 GB per second on a single processor core. A standard server can verify far more traffic than its network card can carry, so this adds no hardware.
- **Decoding the video itself**, for quality checks or server-side processing, runs 6–16 times faster than real time on one core. Per 1,000 hours of content that is **\$1.30–\$3.40 of computing for H.264 or AV1** and **\$2.30–\$5.70 for H.265**. H.265 costs about 70% more to decode, so it is the weakest choice where video is decoded in software.
- **Live video re-packaging**: cameras and studio encoders already send video in a format players can use. The prototype re-wraps a live stream for Apple (HLS) and Android/TV (DASH) players in under a tenth of a second per 15-second segment, about 0.6% of the effort of re-encoding. That saves about 2.2 computer-hours per channel-hour, or **\$32–\$80 per round-the-clock channel per month**. The trade-off is that a re-wrapped stream offers one quality level, not several, so it suits feeds where the source quality is already right for viewers.
- **Unpacking compressed knowledge-base data** runs at 260–410 MB per second, so compression adds no meaningful delay at the receiving end.

Propagation: live video stays fast on a busy network

Giving live video its own priority lane made it 4.3× faster on a congested long-distance link: 0.14 seconds instead of 0.60 seconds. The same link stayed 80% busy with background work, so no extra bandwidth was needed.

- **Without priority lanes**, live video waits in line behind large background transfers such as archive copies and catalogue syncs. Delay rose to 0.6 seconds, enough to put a live feed noticeably behind real time.
- **With priority lanes**, live video went first and background work used whatever capacity was left. The background transfer still moved 16 Mbps of the link's 20 Mbps.
- **What it is worth**: the usual alternative is to reserve capacity for live video and leave it idle the rest of the time. Using that 80% for background work instead is worth **\$810–\$4,050 a month per 10 Gbps** of inter-region capacity, at typical bandwidth prices of

- \$0.10–\$0.50 per Mbps per month. Put another way, about 2,600 TB a month of background transfer rides on capacity already paid for.
- **Live feeds kept pace with real time.** Across 15-second live runs, the transport added no growing backlog. In the one feed that drifted, the delay was already present at the live encoder before transmission, so it is an encoder sizing issue, not a network one.
 - **Recorded video timing tracked distance,** as expected. Each piece took 26 ms regionally, 81 ms long-haul and 42 ms on mobile, within a few milliseconds of each path’s built-in delay. The 13 ms on the office LAN is mostly our test equipment, not distance.

Streaming readiness: works with the players and cameras already in use

The prototype delivered video in all three industry-standard streaming formats, and each copy arrived complete and played back correctly. No new player apps or camera equipment are needed.

Format	Used by	Test result
HLS	iPhone, iPad, Safari, most web players	13 files, two quality levels, complete and playable on arrival
DASH	Android, smart TVs, Chrome-based players	20 files, two quality levels plus audio, complete and playable on arrival
RTMP (live in)	OBS, hardware encoders, cameras	15-second live feed received, then automatically re-packaged as HLS and DASH in under 0.1 seconds each

This lets one ingest point accept live video from standard studio tools and serve it to every major device family. That avoids a separate commercial live-ingest service, typically priced per channel-hour. We have not priced that avoidance here, because vendor rates vary widely.

All streaming files are checked against the playlist that lists them, and against a fingerprint of their contents. A missing or altered segment is caught before any viewer requests it.

Savings model: assumptions to replace with our own figures

Every estimate is a measured result from the tests, multiplied by an illustrative monthly workload and a range of market prices. Each lever scales in a straight line with its volume, so doubling viewer-hours doubles the first two levers.

Illustrative monthly workload

Driver	Assumed volume	Used for
Viewer-hours of video delivered	1,000,000	All delivery savings and overhead
Share watched on slow or mobile connections	30%	Smaller video for slow connections
Share that is fast-moving content (sports, action)	25%	AV1 for fast-moving content
Hours of new fast-moving content encoded	500	AV1 extra encoding cost
Full syncs of the 149M-record knowledge base	30 (daily)	Data compression
Images delivered	10,000,000	Image format
Live channel-hours	2,000 (about 3 channels round the clock)	Live re-packaging
Inter-region capacity shared by live and background traffic	10 Gbps	Priority lanes option
Video library held in archive	10,000 hours	Archive option

Price ranges (low = large-volume contract, high = typical list price)

Cost item	Low	High
Delivering video to viewers (CDN)	\$0.01 per GB	\$0.05 per GB
Moving data between cloud regions	\$0.02 per GB	\$0.09 per GB

Cost item	Low	High
Computing	\$0.02 per core-hour	\$0.05 per core-hour
Network capacity	\$0.10 per Mbps per month	\$0.50 per Mbps per month
Storage	\$0.004 per GB per month (archive tier)	\$0.023 per GB per month (standard tier)

These prices are typical published rates from memory, not quotes. Finance should replace them with our contracted rates. The model, with the measured inputs, is saved alongside the test report in the project repository, so it can be re-run with real figures.

Risks and next steps

The results are strong but come from a controlled lab, so a short pilot on real traffic should confirm them before budget is committed.

What the evidence does not yet cover

- **Real footage.** Test clips were computer-generated with added grain. Real camera footage usually compresses better, so the savings are likely conservative, but this is not yet proven.
- **Lost data on bad networks.** Our test networks added distance and speed limits but never dropped data. Mobile and congested networks do, and that has not been measured.
- **Scale.** Both sites ran on one 4-core machine. The speeds and per-unit costs are reliable, but behaviour at thousands of simultaneous streams is untested.
- **Device support for AV1.** Older phones and TVs cannot play it, so H.264 copies must still be kept, which slightly reduces the net saving.
- **The archive option** depends on whether studio-quality masters must be kept for re-editing or licensing. That is a content-policy decision, not a technical one.

Recommended next steps

- ☐ Finance replaces the illustrative volumes and prices with our real figures and re-runs the model
- ☐ Two-week pilot of the five core levers on a share of real traffic, measuring bills before and after
- ☐ Repeat the network tests with real footage and simulated data loss

- ☐ Check what share of our audience's devices can play AV1
- ☐ Content owners decide on archive policy for older titles